Research Statement

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My research lies at the intersection of deep reinforcement learning, operations research, and optimization, with a strong emphasis on developing efficient algorithms to solve complex, real-world problems. My work mixes deep reinforcement learning with classic optimization and lots of real-world data so the systems make smart decisions in seconds and keep costs down. My research has been published in peer-reviewed A* conferences and deployed in real-world applications within the transportation domain.

My interest in machine learning began when I worked with HL7 messaging protocols used in the healthcare industry. I developed an algorithm to track resource shortages by analyzing admission and registration messages and proposed a way to analyze doctors' diagnoses for early warning of emerging diseases. Later, I explored how natural language processing and data modeling could enhance this work. In my final year, I focused on improving sentiment analysis of tweets using semi-supervised learning. I created a co-training algorithm that outperformed existing methods and combined it with self-training through feature tuning, achieving strong F1-scores.

During my Ph.D., I focused on formulating and solving complex combinatorial optimization problems using deep reinforcement learning (DRL). These included dynamic vehicle routing with advanced requests and real-time emergency responder repositioning, systems characterized by stochasticity, partial observability, and hard temporal constraints. Traditional optimization methods were often too slow for these settings. In contrast, I developed DRL-based agents that learn near-optimal policies via environment interaction and leverage neural approximations for rapid inference during deployment.

My research addressed three core challenges in sequential decision-making under uncertainty: (1) **Proactive Repositioning of Resources**: introduced a DRL formulation using Deep Deterministic Policy Gradient (DDPG) to proactively position responders in anticipation of future demand. The resulting policies replaced computationally expensive tree search methods with transformer-based actor networks that make real-time decisions in large, discrete spaces. (2) **Online Booking with Flexibility**: formulated a model combining two major components (i.e., online booking and offline routing), where I use a trained DRL agent to determine the pickup time at booking and use VRP solvers to optimize the route plans. This system balances efficient utilization of resources while maintaining real-time responsiveness. (3) **Dynamic Routing with Continuous Reoptimization**: devise a model to support advance booking in dynamic routing, where I use simple insertion heuristics to determine the acceptance/rejection of request in real-time and utilize anytime routing to optimize the routes between arrival or requests. The result was significantly improved service coverage in high-demand environments.

Currently, my research focuses on applying probabilistic modeling to integrate multi-modal transportation systems, aiming to enhance accessibility, efficiency, and equity in urban mobility. By combining real-time data, demand forecasting, and optimization techniques, I develop adaptive systems that dynamically coordinate public transit, on-demand shuttles, and micro-mobility solutions. This work is motivated by the need to design transportation networks that are not only cost-effective but also responsive to the evolving needs of diverse communities especially underserved populations.